

Syllabus: STT xxxx - Statistical Learning II

Syllabus - FALL 2025

Instructor: Dr. TBD

Class meeting: TBD

Office: TBA

Online Office Hours: TBD

In-person Office Hours: TBD

Prerequisite: STT 3851

Course Catalog Description:

This course serves as a continuation of STT 3851 (Statistical Learning I). Fundamentals of Generalized Linear Models (GLM) are explored, with a focus on the Linear exponential family, the Link function, and their applications to categorical and count response variables. Supervised learning techniques for classification models and tree-based methods are also examined, as they apply to both regression and classification problems. Unsupervised learning is introduced, including principal component analysis and various clustering algorithms. Other topics such as support vector machines, regression splines, and smoothing splines may be included. Students will complete projects using methods and algorithms covered in class and report their results using professional editing tools that use principles of reproducible statistical research.

Learning Outcomes

At the completion of this course, students will be able to:

- Understand key concepts of statistical learning and the R programming language.
- Explain models in R and access and interpret output, including diagnostics.
- Understand the key concepts concerning the construction of generalized linear models.
- Apply the key concepts concerning principal components analysis.
- Apply the key concepts concerning decision tree models.
- Apply the key concepts concerning cluster analysis.
- Write reports and explain findings related to models and code.

Course Text:

- Introduction to Statistical Learning, by G. James, D. Witten, T. Hastie, and R. Tibshirani (available as pdf download) (referred to as ISLR below) (7th Edition) <http://www-bcf.usc.edu/~gareth/ISL/>
- Regression Modeling with Actuarial and Financial Applications, Edward W. Frees, 2010, New York: Cambridge. ISBN: 978-0521135962

Required Resources:

Text book, Computer with access to the RStudio server (provided), Note book, notes, pencil and a calculator

Homework: Students are expected to work all assigned problems. Only selected assignments or parts of assignments may at times be graded. See schedule for the dates.

Quizzes: I anticipate giving weekly quizzes during the semester. See schedule for the dates.

Tests/ Projects: One mid exam and a project will be given. See schedule for the dates.

Final Exam: A final exam will be given on TBA. This final exam will be comprehensive, covering all material that is covered in the course. As the final exam needs to be proctored you are required to have your video camera on during the final exam time.

Attendance and participation:

Attendance will be taken at the end of each class. Students are expected to attend every class. A detailed description of the ASU attendance policy can be found at the following link, [<https://academicaffairs.appstate.edu/resources/syllabi-policy-and-statement-information>]

Course Grading:

- 20% of the grade will come from Homework.
- 30% of the course grade will come from Quizzes.
- 15% of the grade will come from project.
- 15% of the grade will come from the Mid-Exam.
- 20% of the grade will come from the Final-Exam.

Range	Letter Grade	Range	Letter Grade
92.50% and above	A	72.50-77.49%	C
90.00-92.49%	A-	70.00-72.49%	C-
87.50-89.99%	B+	67.50-69.99%	D+
82.50-87.49%	B	62.50-67.49%	D
80.00-82.49%	B-	60.00-62.49%	D-
77.50-79.99%	C+	59.99% and below	F

Academic Integrity Policy, Disability Services, and Religious Holiday Observance:

All students must abide by the ASU Academic Integrity Code posted at www.studentconduct.appstate.edu/ Those seeking accommodations for a disability should contact the Office of Disability Services (ODS) at <http://www.ods.appstate.edu/> or 828-262-3056. The statements regarding University policies for students are posted at <http://academicaffairs.appstate.edu/syllabi>

University Policies

This course conforms with all Appalachian State University policies with respect to academic integrity, disability services, and class attendance. The details of the policies may be found at <http://academicaffairs.appstate.edu/resources/syllabi>.

Computers and Software

This course will use the RStudio server (<https://mathr.appstate.edu/>) that has the programs listed below and more installed.

- R
- RStudio
- LaTeX

You must have an active internet connection and be registered in the course to access the server. To access the server, point any web browser to <https://mathr.appstate.edu/>. You will need to acknowledge the connection is unsecure and possibly add a security exception to your web browser. Use your Appstate Username and Password to access the server. A screen shot of the RStudio server is shown below.

Sign in to Posit Workbench

Username:

Password:

☐ Stay signed in when browser closes

You will automatically be signed out after 60 minutes of inactivity.

Sign in